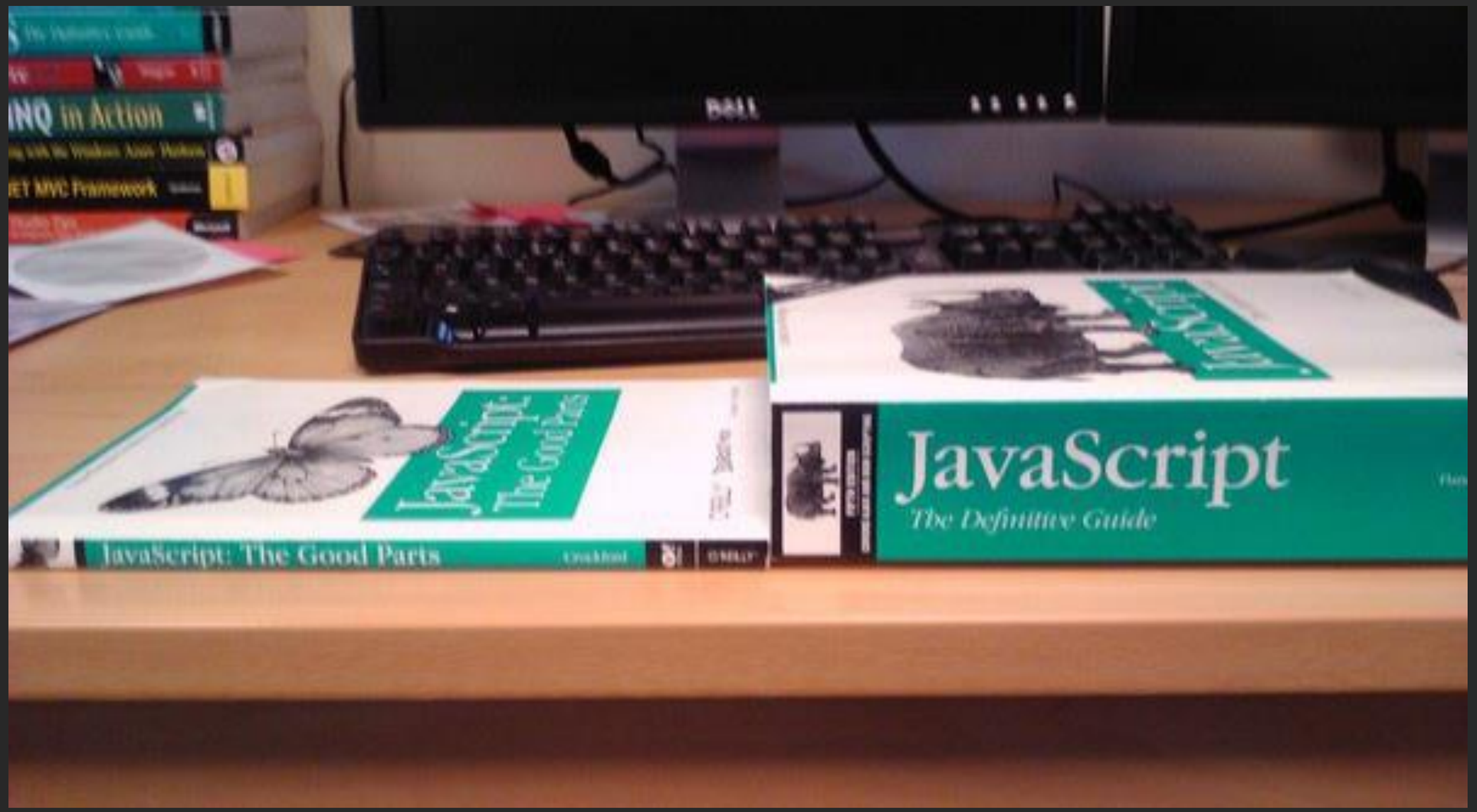
JavaScript Object Notation

- Douglas Crockford - “Discovered” JSON
- Object literal notation in JavaScript

http://www.youtube.com/watch?v=kc8BAR7SHJI


JavaScript
The Definitive Guide
JavaScript: The Good Parts

JSON

http://www.json.org/

Introducing JSON

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JSON (JavaScript Object Notation) is a lightweight data-interchange format. It is easy for humans to read and write. It is easy for machines to parse and generate. It is based on a subset of the JavaScript Programming Language, Standard ECMA-262 3rd Edition - December 1999. JSON is a text format that is completely language independent but uses conventions that are familiar to programmers of the C-family of languages, including C, C++, C#, Java, JavaScript, Perl, Python, and many others. These properties make JSON an ideal data-interchange language.

JSON is built on two structures:

- A collection of name/value pairs. In various languages, this is realized as an *object*, record, struct, dictionary, hash table, keyed list, or associative array.
- An ordered list of values. In most languages, this is realized as an *array*, vector, list, or sequence.

These are universal data structures. Virtually all modern programming languages support them in one form or another. It makes sense that a data format that is interchangeable with programming languages also be based on these structures.

In JSON, they take on these forms:

An *object* is an unordered set of name/value pairs. An object begins with { (left brace) and ends with } (right

object
{}
{ *members* }
members
pair
pair , *members*
pair
string : *value*
array
[]
[*elements*]
elements
value
value , *elements*
value
string
number
object

```
import json
data = '''{
  "name" : "Chuck",
  "phone" : {
    "type" : "intl",
    "number" : "+1 734 303 4456"
   },
   "email" : {
     "hide" : "yes"
   }
}'''

info = json.loads(data)
print('Name:',info["name"])
print('Hide:',info["email"]["hide"])
```

json1.py

JSON represents data as nested “lists” and “dictionaries”

json2.py

```
import json
input = '''[
  { "id" : "001",
    "x" : "2",
    "name" : "Chuck"
  } ,
  { "id" : "009",
    "x" : "7",
    "name" : "Chuck"
  }
]'''

info = json.loads(input)
print('User count:', len(info))
for item in info:
    print('Name', item['name'])
    print('Id', item['id'])
    print('Attribute', item['x'])
```

JSON represents data as nested “lists” and “dictionaries”

Service Oriented Approach

http://en.wikipedia.org/wiki/Service-oriented_architecture

Acknowledgements / Contributions

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